APPSC INTEGRATED SUBJECT INDEX & MICRO-TOPIC OVERLAP ANALYZER

06. ENVIRONMENTAL SCIENCE & ENGINEERING

Underlined Highlights: AEE PR Extras | AEE PCB Extras | Group-1/2 Extras

GR-2 MAINS: P-II S&T (Units 2–5) | AEE PR/TW: P-III (Units 2–8) | AEE PCB: P-II (Units 1–3, 7–10) | AEE CIVIL: P-III (Unit 7) | GR-1: PRELIMS + MAINS P-V (Units 4–6)

Unit	Consolidated Theme Title	Exams & Notification Clause Mapping	Overlap Type	Page
01	**Ecosystem Structure & Functions**	GR2 S&T U3 • AEE PR U3 • AEE PCB U1 • AEE Civil U7	High Common + PCB Extras	01
02	**Biodiversity & Biotic Resources**	GR2 S&T U3 • AEE PR U4 • AEE PCB U3 • GR1 P-V U6	90% Common + GR2/PCB Acts	02
03	**Natural Resources & Energy Mgmt**	GR2 S&T U2 • AEE PR U2 • AEE PCB U2 • GR1 P-V U4,5	PR Social Extras + GR2 Energy	02–03
04	**Environmental Pollution & Control Tech**	GR2 S&T U4 • AEE PR U4 • AEE PCB U7 • AEE Civil U7	Concept (PR/GR2) vs Design (PCB/Civ)	03–04
05	**Solid, Industrial & E-Waste Mgmt**	GR2 S&T U4 • AEE PR U5 • AEE PCB U8 • AEE Civil U7	Core MSW + PCB E-Waste/Biomed	04
06	**Climate Change & Global Env Issues**	GR2 S&T U5 • AEE PR U7 • AEE PCB U9 • GR1 P-V U6	Core Global + PR Social Extras	05
07	**Env Policies, Conventions & Laws**	GR2 S&T U5 • AEE PR U7 • AEE PCB U9 • GR1 P-V U6	5 Core Acts + PCB Handling Rules	05
08	**EIA, Sustainability & Social Issues**	GR2 S&T U5 • AEE PR U7 • AEE PCB U10 • AEE Civil U7	SDGs (GR2) + R&R (PR) + EIA/LCA (PCB)	06
09	**Human Population, Health & Env**	GR2 S&T U5 • AEE PR U8 • AEE PCB U10 • GR1 P-V U8	Demography/Welfare (PR) + Diseases (GR2)	06
10	**Disaster Management & GS-150 Line**	AEE PR U6 • Common GS P-I (Items 7,8) • Draughtsman P-I U8	Hazards (PR) + RS/GIS Profile (GS-150)	06

UNIT 01 • ECOSYSTEM STRUCTURE, FUNCTION & DYNAMICS

GR2 P-II U3 | AEE PR P-III U3 | AEE PCB P-II U1 | AEE CIVIL P-III U7

3. Ecosystem and Biodiversity — GR2 MAINS P-II S&T

Ecology and Ecosystem: Basic concepts of Ecology, Ecosystem: Components and Types.

3. ECOSYSTEMS — AEE PR/TW PAPER-III

Concept of an ecosystem. – Structure and function of an ecosystem. – Producers, consumers and decomposers. – Energy flow in the ecosystem – Ecological succession. – Food chains, food webs and ecological pyramids. – Introduction, types and characteristic features, Structure and function of Forest, Grassland, Desert and Aquatic ecosystems.

1. ECOSYSTEMS — AEE PCB PAPER-II

Definition, Scope and Importance of ecosystem. Classification, structure and function of an ecosystem, Food chains, food webs and ecological pyramids. Flow of energy, Biogeochemical cycles, Bioaccumulation, Biomagnification, ecosystem value, services and carrying capacity.

100% Overlapping Sub-Topics (Read Once)	Extra Sub-Topics ONLY in AEE PR/TW Paper-III	Extra Sub-Topics ONLY in AEE PCB Paper-II
• Definition, scope & importance of Ecology • Abiotic & Biotic components	• Ecological Succession (Primary/Secondary, Hydrarch/Xerarch	• Biogeochemical Cycles (Carbon, Nitrogen, P, S)

UNIT 02 • BIODIVERSITY & BIOTIC RESOURCES

GR2 P-II U3 | AEE PR P-III U4 | AEE PCB P-II U3 | GR1 P-V U6,7

3. Ecosystem and Biodiversity

GR2 MAINS P-II S&T

Biodiversity: Meaning, Components, Biodiversity Hotspots, Loss of Biodiversity and Conservation of Biodiversity: Methods, Recent Plans, Targets, Convention and Protocols – Wildlife Conservation: CITES and Endangered Species with reference to India – Biosphere Reserves – Indian Wildlife Conservation efforts, projects, acts and initiatives in recent times.

4. BIODIVERSITY AND ITS CONSERVATION

AEE PR/TW PAPER-III

Introduction – Definition: genetic, species and ecosystem diversity. – Bio-geographical classification of India – Value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values – Biodiversity at global, national and local levels. – India as a mega-diversity nation – Hot-spots of biodiversity – Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts. – Endangered and endemic species of India – Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.

3. BIODIVERSITY AND BIOTIC RESOURCES

AEE PCB PAPER-II

Introduction, Definition, genetic, species and ecosystem diversity. Value of biodiversity... Hot spots of biodiversity. Field visit. Threats to biodiversity... conservation of biodiversity... National Biodiversity act.

100% Overlapping Sub-Topics	Extra ONLY in AEE PR/TW	Extra ONLY in AEE PCB	Extra ONLY in Gr-2 / Gr-1
• Genetic, Species & Ecosystem diversity • 6 Values of Biodiversity • India Hotspots & Threats • In-situ vs Ex-situ conservation	• Bio-geographical Classification (10 Zones) • Endemic Species lists	• National Biodiversity Act, 2002 • Field visit documentation	• Kunming-Montreal Framework (30x30) • CITES & Species Projects

UNIT 03 • ENVIRONMENTAL SCIENCE, NATURAL RESOURCES & ENERGY

GR2 P-II U2 | AEE PR P-III U2 | AEE PCB P-II U2 | GR1 P-V U4,5

2. Energy Management

GR2 MAINS P-II S&T

Policy and Projections: Installed Energy Capacities and Demand in India – National Energy Policy – National Policy on Biofuels – Bharat Stage Norms – Non-Renewable and Renewable Energy: Sources and Installed Capacities in India – New Initiatives and Recent Programmes, Schemes and Achievements in India's Renewable Energy Sector.

2. Environmental science (Natural Resources)

AEE PR/TW PAPER-III

Definition, Scope and Importance... Forest resources: Use and over exploitation, deforestation, Timber extraction, Mining, dams and other effects on forest and tribal people – Water resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity – Energy resources... Land resources: Role of an individual in conservation of natural resources. Equitable use of resources for sustainable lifestyles.

Unit 03: 100% Common Core (PR + PCB + GR1)	Extra Sub-Topics ONLY in AEE PR/TW Paper-III (Absent in PCB!)	Extra Sub-Topics ONLY in Group-2 S&T & Group-1 P-V
• Renewable/Non-renewable resources • Water, Mineral, Forest & Land basic uses • Growing energy needs & alternate sources	• Forest & Tribal Impact (Mining/Dams on tribal people) • Inter-State/International Water Conflicts • Food Resources & Agro-chemical/Salinity issues • Land Hazards (Man-induced landslides, soil erosion, desertification) • Individual Conservation Role & Sustainable Lifestyles	• Installed Energy Capacities in India & AP • National Energy Policy & Biofuels Policy • Bharat Stage (BS-VI) Vehicular Norms • MNRE Schemes (KUSUM, Surya Ghar, Green Hydrogen)

UNIT 04 • ENVIRONMENTAL POLLUTION & CONTROL TECHNOLOGIES

GR2 P-II U4 | AEE PR P-III U4 | AEE PCB P-II U7 | AEE CIVIL P-III U7

4. Waste Management and Pollution Control — GR2 MAINS P-II S&T

Environmental Pollution: Types, Sources and Impacts... Impact of Transgenics on Environment and their Regulation – Eco-Friendly Technologies in Agriculture – Bioremediation: Types and Scope in India.

4. ENVIRONMENTAL POLLUTION — AEE PR/TW PAPER-III

Definition, Cause, effects and control measures of Air pollution, Water pollution, Soil pollution, Marine pollution, Noise pollution, Thermal pollution and Nuclear hazards.

7. AIR, WATER, SOIL, NOISE POLLUTION & CONTROL TECHNOLOGIES — AEE PCB PAPER-II

Classification... Air Pollution: Primary/secondary pollutants, Particulate emission control, SOx/NOx control, Automobile/Industrial pollution, NAAQS. Water pollution: Drinking water standards, Wastewater Treatment (Grit chambers, Sedimentation, Trickling filters, Oxidation ponds, ASP, Septic tanks, Sludge disposal, Recycling). Soil & Noise standards.

Unit 04: Common Core	Extra ONLY in AEE PR & Gr-2	Extra ONLY in AEE PCB Paper-II	Extra ONLY in AEE Civil Paper-III
• Primary/Secondary pollutants • Water & Soil pollution sources • Noise decibel standards & NAAQS	• Marine, Thermal & Nuclear hazards • Transgenics (GMOs) & GEAC regulation • Eco-Friendly Agriculture & Bioremediation	• Air control hardware (ESP, Scrubbers, Cyclones, Baghouse) • Wastewater unit processes (ASP, Trickling filters, Oxidation ponds)	• Raw Water supply & distribution networks (Hardy-Cross) • Storm water drainage & Sewerage hydraulics (Self-cleansing velocity)

UNIT 05 • SOLID, INDUSTRIAL & E-WASTE MANAGEMENT

GR2 P-II U4 | AEE PR P-III U5 | AEE PCB P-II U8 | AEE CIVIL P-III U7

5. SOLID WASTE MANAGEMENT — AEE PR/TW PAPER-III

Causes, effects and control measures of urban and industrial wastes. Role of an individual in prevention of pollution.

8. SOLID AND INDUSTRIAL WASTE MANAGEMENT — AEE PCB PAPER-II

Municipal Solid Waste management, Collection, Segregation and Transport, Solid waste processing technologies, composition and characteristics of e-Waste and its management, Concepts of bioremediation.

Unit 05: Common Core	Extra ONLY in AEE PR/TW	Extra ONLY in AEE PCB & Group-2
• MSW Classification • SWM Hierarchy (3Rs) • Sanitary Landfill & Composting	• Role of an individual in prevention of pollution	• E-Waste heavy metals & E-Waste Rules 2022 • Collection, Segregation & Transport mechanics (Haul/Stationary container systems) • Bioremediation (In-situ & Ex-situ techniques)

UNIT 06 • CLIMATE CHANGE & GLOBAL ENVIRONMENTAL ISSUES

GR2 P-II U5 | AEE PR P-III U7 | AEE PCB P-II U9 | GR1 P-V U5,6

5. Environment and Health (Environment Challenges) — GR2 MAINS P-II S&T

Environment Challenges: Global Warming, Climate Change, Acid Rain, Ozone Layer Depletion, Ocean Acidification.

7. SOCIAL ISSUES AND THE ENVIRONMENT (Global Problems) — AEE PR/TW PAPER-III

Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Wasteland reclamation. Consumerism and waste products.

9. GLOBAL ENVIRONMENTAL PROBLEMS — AEE PCB PAPER-II

Climate change and impacts on human environment. Ozone depletion and Ozone depleting substances (ODS). Deforestation and desertification.

Unit 06: Common Core	Extra ONLY in Group-1/2	Extra ONLY in AEE PR/TW	Extra ONLY in AEE PCB
• Greenhouse Gases & GWP • Ozone layer depletion mechanism • Acid Rain chemistry	• Ocean Acidification • Climate Justice & Net-Zero 2070	• Nuclear Accidents & Holocaust • Wasteland Reclamation • Consumerism & Waste products	• ODP Ratings of ODS & Kigali Amendment • Deforestation & Desertification linkages

UNIT 07 • ENVIRONMENTAL POLICIES, CONVENTIONS & LEGISLATIONS

GR2 P-II U5 | AEE PR P-III U7 | AEE PCB P-II U9 | GR1 P-V U6

7. SOCIAL ISSUES AND THE ENVIRONMENT (Legislation) — AEE PR/TW PAPER-III

Environment Protection Act. Air Act. Water Act. Wildlife Protection Act. Forest Conservation Act – Issues involved in enforcement of environmental legislation. Public awareness.

9. GLOBAL ENVIRONMENTAL PROBLEMS & LEGISLATIONS — AEE PCB PAPER-II

International conventions / Protocols: Earth summit, Kyoto protocol, Montréal Protocol and Paris convention, Indian EP Act, Legal aspects Air Act, Water Act, Forest Act, Wild life Act, Municipal solid waste management rules, biomedical waste management rules, hazardous waste management rules.

Unit 07: Common Core (5 Acts + Rio/Kyoto/Paris)	Extra ONLY in AEE PR & Group-2	Extra ONLY in AEE PCB Paper-II
• 5 Core Indian Acts (1972, 1974, 1980, 1981, 1986) • Rio Summit, UNFCCC, Kyoto, Montreal, Paris	• Enforcement issues & Public awareness • Recent COP Summits (COP28/29/30), ISA, Mission LiFE	• Solid Waste Management Rules 2016 • Bio-Medical Waste Rules 2016 (4 Bins) • Hazardous Wastes Rules 2016 (Manifest system)

UNIT 08 • EIA, SUSTAINABLE DEVELOPMENT & SOCIAL ISSUES

GR2 P-II U5 | AEE PR P-III U7 | AEE PCB P-II U10 | AEE CIVIL P-III U7

7. SOCIAL ISSUES AND THE ENVIRONMENT

AEE PR/TW PAPER-III

From Unsustainable to Sustainable development – Urban problems related to energy – Water conservation, rain water harvesting, watershed management – Resettlement and rehabilitation of people; its problems and concerns. – Environmental ethics .

10. ENVIRONMENTAL IMPACT ASSESSMENT

AEE PCB PAPER-II

EIA structure, baseline data acquisition, Air/Water/Biological/Socio-economic impacts, Risk assessment, EMP, Concept of Sustainable Development, Crazy Consumerism, Environmental Education, Urban Sprawl, Human health, Green Building, Ecological Foot Print, Life Cycle assessment (LCA), Low carbon life style .

Unit 08: Common Core	Extra ONLY in Group-2	Extra ONLY in AEE PR/TW	Extra ONLY in AEE PCB
• Sustainable Development concept • Environmental ethics • EIA basics	• UN 17 SDGs & NITI Aayog SDG Index	• Rainwater harvesting & Watershed mgmt • Resettlement & Rehabilitation (R&R) • Urban energy problems	• Technical EIA (Leopold matrix, EMP) • Green Buildings (GRIHA/LEED) • Ecological Footprint & LCA

UNITS 09 & 10 • HUMAN POPULATION, HEALTH & DISASTER MANAGEMENT

GR2 P-II U5 | AEE PR P-III U6,8 | AEE PCB P-II U10 | COMMON GS P-I

8. HUMAN POPULATION & 6. DISASTER MANAGEMENT

AEE PR/TW PAPER-III

Population growth, variation among nations. Population explosion – Family Welfare Programme . Environment and human health. Human Rights. Value Education. HIV/AIDS. Women and Child Welfare. Role of IT in Environment and human health . **Unit 6:** Disaster management: floods, earthquake, cyclone and landslides.

COMMON GS PAPER-I (150M) & DRAUGHTSMAN PAPER-I

GS & DRAUGHTSMAN

Sustainable Development and Environmental Protection. Disaster management: vulnerability profile, prevention and mitigation strategies, Application of Remote Sensing and GIS in disaster assessment . (Draughtsman adds: Tsunami, Drought & Manmade disasters) .

Units 09–10: Common Core	Extra ONLY in AEE PR/TW (U8)	Extra ONLY in Group-1/2	Extra ONLY in Common GS & DM
• Population curves & explosion • Environment & human health • Floods, Earthquakes, Cyclones	• Family Welfare Programme • Human Rights & Value Education • HIV/AIDS & Women/Child Welfare • Role of IT in Env & Health	• Disease Burden & Pandemic Preparedness • Biotech/Nanotech infections (Gr-1)	• DM Act 2005 & Sendai Framework • Remote Sensing & GIS in Disasters • Manmade disasters (Draughtsman)